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Laminated glass and head-up display system

Abstract

The disclosure relates to a laminated glass and a head-up display system. The laminated glass includes a first transparent substrate, a second transparent substrate, and an adhesive film. The laminated glass has a light-transmitting region and a light-blocking region surrounding at least part of a periphery of the light-transmitting region. The adhesive film is disposed between the first transparent substrate and the first transparent substrate and configured to adhere the first transparent substrate and the second transparent substrate. The light-transmitting region has a visible light transmittance greater than or equal to 70%. The light-blocking region has a visible light transmittance less than or equal to 5%. The light-blocking region has a first region located below the light-transmitting region, and the first region has one or more first function display regions for displaying of image.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of International Application No. PCT/CN2022/123855, filed on Oct. 8, 2022, which claims priority to Chinese Patent application Ser. No. 20/211,1173403.2, filed Oct. 8, 2021, and Chinese Patent Application No. 202111173404.7, filed Oct. 8, 2021, the entire disclosures of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD
(1) The disclosure relates to the field of automobiles, and in particular, to a laminated glass and a head-up display system.
BACKGROUND
(2) With the development of automotive intelligence, head-up display (HUD) systems are increasingly applied to automobiles to project images, such as real-time driving information, onto a front windshield. Due to the front windshield being laminated glass, a light ray emitted by a projection light source of the HUD system will undergo reflection when passing through each of

two surfaces of the laminated glass in contact with air. The reflection on each of the two surfaces causes deviation, resulting in two mutually interfering ghost images, thus leading to a decrease in the quality of the image projected onto the front windshield.

SUMMARY

(3) A laminated glass is provided in the disclosure. The laminated glass includes a first transparent substrate, a second transparent substrate, and an adhesive film. The first transparent substrate has a first surface and a second surface opposite the first surface. The second transparent substrate has a third surface and a fourth surface opposite the third surface. The third surface is closer to the second surface than the fourth surface. The laminated glass has a light-transmitting region and a light-blocking region surrounding at least part of a periphery of the light-transmitting region. The adhesive film is disposed between the second surface and the third surface and configured to adhere the first transparent substrate and the second transparent substrate. The light-transmitting region has a visible light transmittance greater than or equal to 70%. The light-blocking region has a visible light transmittance less than or equal to 5%. The light-blocking region has a first region located below the light-transmitting region, and the first region has one or more first function display regions for displaying of a first image.

(4) In some embodiments, the light-blocking region further has a second region located above the light-transmitting region and a third region disposed at two sides of the light-transmitting region.

(5) In some embodiments, at least one flexible display screen is disposed in the first function display region. The at least one flexible display screen is disposed between the second surface and the third surface and includes at least one of a mini light-emitting diode (LED) display screen, a micro LED display screen, or an organic light-emitting diode (OLED) display screen.

(6) In some embodiments, the first function display region is configured to receive a projection light ray for forming the first image that is incident at an angle of 50° to 72°. The first function display region has a reflectivity for the projection light ray incident greater than or equal to 4%.

(7) In some embodiments, the first function display region is part of the fourth surface, a proportion of S-polarized light in the projection light ray incident ranges from 60% to 100%. The first function display region has the reflectivity for the projection light ray incident greater than or equal to 8%.

(8) In some embodiments, the laminated glass further includes a dielectric film disposed in the first function display region, the dielectric film is on the third surface or the fourth surface. A proportion of P-polarized light in the projection light ray incident ranges from 60% to 100%, and the first function display region has the reflectivity for the projection light ray incident greater than or equal to 8%. Alternatively, a proportion of S-polarized light in the projection light ray incident ranges from 60% to 100%, and the first function display region has the reflectivity for the projection light ray incident greater than or equal to 8%.

(9) In some embodiments, the laminated glass further includes a metal film disposed in the first function display region. The metal film is on the third surface. A proportion of P-polarized light in the projection light ray incident ranges from 60% to 100%. The first function display region has the reflectivity for the projection light ray incident greater than or equal to 6%.

(10) In some embodiments, the laminated glass further includes a laminated polyethylene terephthalate (PET) disposed in the first function display region, a proportion of P-polarized light in the projection light ray incident ranges from 60% to 100%. The first function display region has the reflectivity for the projection light ray incident greater than or equal to 10%.

(11) In some embodiments, the laminated glass further includes a dark ink layer or a colored polymer film disposed in the light-blocking region. The dark ink layer is disposed on at least one of the second surface or the third surface. The colored polymer film is disposed between the second surface and the third surface.

(12) In some embodiments, the flexible display screen or the display region is closer to the fourth surface than the dark ink layer or the colored polymer film.

(13) In some embodiments, the fourth surface has a colored region. An upper boundary of the first

region is at least 80 mm higher than an upper boundary of the colored region located in the first region.
(14) In some embodiments, the light-transmitting region has one or more second function display regions. The one or more second function display regions are configured for displaying of a second image.
(15) In some embodiments, a projection display distance for the first image ranges from 0.5 m to 5 m, and a projection display distance for the second image is greater than or equal to 7.5 m.
(16) In some embodiments, a projection light ray for forming the first image is incident on the one or more first function display regions at an angle of 50° to 72°. The one or more first function display regions have a reflectivity, for the projection light ray for forming the first image, greater than or equal to 4%. A projection light ray for forming the second image is incident on the one or more second function display regions at an angle of 50° to 72°. The one or more second function display regions have a reflectivity, for the projection light ray for forming the second image, greater than or equal to 8%.
(17) In some embodiments, the laminated glass further includes a dielectric film. The dielectric film is at least located in the one or more second function display regions.
(18) In some embodiments, the dielectric film is further located in the one or more first function display regions.
(19) In some embodiments, the adhesive film is of uniform thickness. A proportion of P-polarized light in the projection light ray for forming the second image ranges from 60% to 100%. The dielectric film is a laminated structure formed by a high refractive index layer and a low refractive index layer stacked with each other or includes at least one metal layer or laminated PET. The one or more second function display regions have the reflectivity, for the projection light ray for forming the second image incident at an angle of 50° to 72°, greater than or equal to 10%.
(20) In some embodiments, the adhesive film is of uniform thickness or wedge-shaped. The dielectric film is disposed on the fourth surface and is an anti-reflective film. The second function display region is part of the first surface. A proportion of S-polarized light in the projection light ray for forming the second image ranges from 60% to 100%. The anti-reflective film has a reflectivity for the projection light ray for forming the second image less than or equal to 6%. The one or more second function display regions have the reflectivity, for the projection light ray for forming the second image incident at an angle of 50° to 72°, greater than or equal to 8%.
(21) In some embodiments, the adhesive film is wedge-shaped, a proportion of S-polarized light in the projection light ray for forming the second image ranges from 60% to 100%. The dielectric film is a laminated structure disposed on the third surface or the fourth surface and formed by a high refractive index layer and a low refractive index layer stacked with each other. The one or more second function display regions have the reflectivity, for the projection light ray for forming the second image incident at an angle of 50° to 72°, greater than or equal to 28%.
(22) In some embodiments, the adhesive film is wedge-shaped, the second function display region is part of the fourth surface, a proportion of S-polarized light in the projection light ray for forming the second image ranges from 60% to 100%. The one or more second function display regions have the reflectivity, for the projection light ray for forming the second image incident at the angle of 50° to 72°, greater than or equal to 8%.
(23) In some embodiments, a proportion of S-polarized light in a projection light ray for forming the first image ranges from 60% to 100%, or a proportion of P-polarized light in the projection light ray for forming the first image ranges from 60% to 100%.
(24) In some embodiments, the light-transmitting region further has a primary viewing region. The one or more second function display regions are disposed within the primary viewing region, and a lower boundary of the primary viewing region is at least 25 mm higher than an upper boundary of the first region.
(25) A head-up display system is further provided in the disclosure. The head-up display system

includes a first projection light source and the above-mentioned laminated glass. The first projection light source is configured to emit a projection light ray for forming the first image to the first function display region.
(26) In some embodiments, the light-transmitting region has one or more second function display regions. The head-up display system further includes a second projection light source. The second projection light source is configured to emit a projection light ray for forming a second image to the second function display region.
(27) In some embodiments, a proportion of P-polarized light in the projection light ray for forming the first image ranges from 60% to 100%, and a proportion of S-polarized light in the projection light ray for forming the second image ranges from 60% to 100%.
(28) In some embodiments, a proportion of S-polarized light in the projection light ray for forming the first image ranges from 60% to 100%, and a proportion of P-polarized light in the projection light ray for forming the second image ranges from 60% to 100%.
(29) In some embodiments, a proportion of P-polarized light in the projection light ray for forming the first image ranges from 60% to 100%, and a proportion of P-polarized light in the projection light ray for forming the second image ranges from 60% to 100%.
(30) In some embodiments, a proportion of S-polarized light in the projection light ray for forming the first image ranges from 60% to 100%, and a proportion of S-polarized light in the projection light ray for forming the second image ranges from 60% to 100%.
(31) The laminated glass provided in the embodiments of the disclosure has the light-blocking region, which can reduce or even block reflected light A, where light incident on the fourth surface can enter the laminated glass and then be reflected by the first transparent substrate to form the reflected light A, thereby weakening or even blocking ghosting caused by the reflected light A and the reflected light B, where light incident on the fourth surface can enter the laminated glass and then be reflected by the second transparent substrate to form the reflected light B. Additionally, the laminated glass provided in the embodiments of the disclosure can reduce or even block incident light C which is incident on the first surface to enter the laminated glass, thereby weakening or even blocking ghosting caused by the reflected light B and the incident light C. Therefore, it can be seen that the laminated glass provided in the embodiments of the disclosure can enhance the quality of images projected thereon.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic structural view illustrating region division of a laminated glass provided in an embodiment of the disclosure.
(2) FIG. **2** is a cross-sectional view of the laminated glass taken along line I-I of FIG. **1** according to an embodiment of the disclosure.
(3) FIG. **3** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **1** according to an embodiment of the disclosure.
(4) FIG. **4** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **1** according to another embodiment of the disclosure.
(5) FIG. **5** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **1** according to another embodiment of the disclosure.
(6) FIG. **6** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **1** according to another embodiment of the disclosure.
(7) FIG. **7** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **1** according to another embodiment of the disclosure.
(8) FIG. **8** is a schematic structural view illustrating region division of a laminated glass provided

in another embodiment of the disclosure.
(9) FIG. **9** is a schematic structural view illustrating region division of a laminated glass provided in another embodiment of the disclosure.
(10) FIG. **10** is a schematic structural view illustrating region division of a laminated glass provided in another embodiment of the disclosure.
(11) FIG. **11** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure.
(12) FIG. **12** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure.
(13) FIG. **13** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure.
(14) FIG. **14** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure.
(15) FIG. **15** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure.
(16) FIG. **16** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure.
(17) FIG. **17** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure.
(18) FIG. **18** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure.
(19) FIG. **19** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure.
(20) FIG. **20** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure.
(21) FIG. **21** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure.
(22) FIG. **22** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure.
(23) FIG. **23** is a schematic structural view illustrating region division of a laminated glass provided in another embodiment of the disclosure.
(24) FIG. **24** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **23** according to another embodiment of the disclosure.
(25) FIG. **25** is a schematic structural view illustrating region division of a laminated glass provided in another embodiment of the disclosure.
(26) FIG. **26** is a schematic structural view illustrating region division of a laminated glass provided in another embodiment of the disclosure.
(27) FIG. **27** is a schematic view of a vehicle provided in the disclosure.
(28) Reference numbers are described as follows: laminated glass **10**, first transparent substrate **110**, first surface **111**, second surface **112**, second transparent substrate **120**, third surface **121**, fourth surface **122**, light-blocking region R**10**, light-transmitting region R**20**, adhesive film **130**, masking layer **140**, first region R**110**, first function display region R**111**, first image P**1**, second region R**120**, third region R**130**, dielectric film **150**, flexible display screen **160**, first projection light source **170**, primary viewing region R**210**, second function display region R**211**, second image P**2**, colored region R**30**, colored layer **180**, second projection light source **190**, vehicle **1**, vehicle body **20**.

DETAILED DESCRIPTION

(29) The technical solutions in the embodiments of the disclosure are clearly and completely described in the following with reference to the accompanying drawings in the embodiments of the disclosure. Apparently, the described embodiments are merely a part of rather than all the

embodiments of the disclosure. All other embodiments obtained by those of ordinary skill in the art based on the embodiments of the disclosure without creative efforts are within the scope of the disclosure.

(30) The term “embodiment” or “example” referred to herein means that a particular feature, structure, or feature described in connection with the embodiment or example may be contained in at least one embodiment of the disclosure. The phrase appearing in various places in the specification does not necessarily refer to the same embodiment, nor does it refer an independent or alternative embodiment that is mutually exclusive with other embodiments. It is expressly and implicitly understood by those skilled in the art that an embodiment described herein may be combined with other embodiments.

(31) Referring to FIGS. **1** to **3**, FIG. **1** is a schematic structural view illustrating region division of a laminated glass provided in an embodiment of the disclosure, FIG. **2** is a cross-sectional view of the laminated glass taken along line I-I od FIG. **1** according to an embodiment of the disclosure, and FIG. **3** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **1** according to an embodiment of the disclosure. A laminated glass **10** is provided in the disclosure. The laminated glass **10** includes a first transparent substrate **110**, a second transparent substrate **120**, and an adhesive film **130**. The first transparent substrate **110** has a first surface **111** and a second surface **112** opposite the first surface **111**. The second transparent substrate **120** has a third surface **121** and a fourth surface **122** opposite the third surface **121**. The third surface **121** is closer to the second surface **112** than the fourth surface **122**. The laminated glass **10** has a light-transmitting region R**20** and a light-blocking region R**10** surrounding at least part of a periphery of the light-transmitting region R**20**. The adhesive film **130** is disposed between the second surface **112** and the third surface **121** and configured to adhere the first transparent substrate **110** and the second transparent substrate **120**. The light-transmitting region R**20** has a visible light transmittance greater than or equal to 70%. The light-blocking region R**10** has a visible light transmittance less than or equal to 5%. The light-blocking region R**10** has a first region R**110** located below the light-transmitting region R**20**, and the first region R**110** has one or more first function display regions R**111** for displaying of a first image P**1**.

(32) In an embodiment, the first transparent substrate **110** is in close connection with the second transparent substrate **120** via the adhesive film **130**. In order to conveniently and clearly illustrate a layered structure of the laminated glass **10**, in the disclosure, a cross-sectional view of FIG. **1** taken along line I-I is rotated by 90 degrees counterclockwise, all the structures of the laminated glass **10** are separated, and the thickness of all the structures is amplified. For the convenience of illustration, the resulting figured is referred to as an exploded cross-sectional structural view taken along line I-I of FIG. **1**. For example, referring to FIG. **3**, FIG. **3** is an exploded cross-sectional structural view taken along line I-I of FIG. **1** according to an embodiment of the disclosure. FIG. **3** is obtained by rotating FIG. **2** by 90 degrees counterclockwise, separating all structures of the laminated glass **10**, and amplifying the thickness of all structures. It can be understood that, the subsequent exploded cross-sectional structural views are also illustrated with reference to the processing manners of FIGS. **2** and **3**, and will not be described in detail hereinafter.

(33) Each of the first transparent substrate **110** and the second transparent substrate **120** may be a curved plate with light-transmitting capabilities. For example, each of the first transparent substrate **110** and the second transparent substrate **120** may be made of inorganic glass or organic glass. The inorganic glass may be, for example, soda-lime-silica glass, aluminum silicate glass, lithium aluminum silicate glass, or borosilicate glass. The organic glass may be, for example, polycarbonate (PC) glass, polymethyl methacrylate (PMMA) glass, or the like. Each of the first transparent substrate **110** and the second transparent substrate **120** may be transparent, or may be colored and have light-transmitting capabilities. The material of the first transparent substrate **110** may be the same as or different from the material of the second transparent substrate **120**.

(34) The light-transmitting region R**20** is a region of the laminated glass **10** through which visible

light can pass. In order to ensure driving safety after the laminated glass **10** is mounted to a vehicle, a visible light transmittance of the light-transmitting region R**20** is preferably greater than or equal to 70%. The light-blocking region R**10** refers to a region of the laminated glass **10** where a visible light transmittance is relatively low. The light-blocking region R**10** is disposed around a periphery of the laminated glass **10**.

(35) The adhesive film **130** is disposed between the first transparent substrate **110** and the second transparent substrate **120** to adhere the first transparent substrate **110** and the second transparent substrate **120**. The adhesive film **130** may be implemented in two structures, which will be described in detail hereinafter.

(36) The laminated glass **10** further includes a masking layer **140** disposed in the light blocking region R**10**. The masking layer **140** may be a dark ink layer or a colored polymer film. The dark ink layer is disposed on at least one of the second surface **112** or the third surface **121**. The colored polymer film is disposed between the second surface **112** and the third surface **121**. The masking layer **140** has a low transmittance for a projection light ray. The masking layer **140** is carried on the first transparent substrate **110** or the second transparent substrate **120** and located in the light-blocking region R**10**. The masking layer **140** may be formed in the light-blocking region R**10** through processes such as printing ink. Optionally, the masking layer **140** has a transmittance for the projection light ray less than or equal to 5%, preferably less than or equal to 1%. Optionally, the masking layer **140** may also be made of a resin film with dark color and low light transmittance. Alternatively, the masking layer **140** may also be made of a resin film with light color and low light transmittance. The resin film may be, for example, a polyvinyl butyral (PVB) or polyethylene terephthalate (PET) which is mass colored.

(37) In an embodiment, referring to FIG. **3**, FIG. **3** is an exploded cross-sectional structural view of the laminated glass **10** taken along line I-I of FIG. **1** according to one embodiment of the disclosure. The masking layer **140** is disposed on the second surface **112**. In another embodiment of the disclosure, referring to FIG. **4**, FIG. **4** is an exploded cross-sectional structural view taken along line I-I of FIG. **1** according to another embodiment of the disclosure. The masking layer **140** is disposed on the third surface **121**.

(38) Specifically, if the laminated glass **10** does not include the masking layer **140**, a projection light ray emitted by a projection device in the vehicle, for forming the first image P**1** onto the laminated glass **10**, is incident on the fourth surface **122** to enter the laminated glass **10**, and then will be reflected by the first surface **111** of the first transparent substrate **110** to form reflected light A. Accordingly, the projection light ray incident onto the laminated glass **10** will also be reflected by the fourth surface **122** of the second transparent substrate **120** to enter the human eye. For the convenience of illustration, the projection light ray reflected by the fourth surface **122** is referred to as reflected light B. The reflected light B forms a main image visible to the human eye. The reflected light A forms a secondary image visible to the human eye. There is a certain offset between the secondary image and the main image. That is, a phenomenon of ghosting occurs. In the embodiments of the disclosure, the laminated glass **10** includes the masking layer **140**, which can reduce or even block the reflected light A, thereby weakening or even blocking the ghosting caused by the reflected light A and the reflected light B. In addition, since the masking layer **140** has a low transmittance for the projection light ray, the masking layer **140** can serve as a display background for the main image, thereby improving the visibility of the main image and the contrast in brightness between the main image and the ambient, and significantly improving the display quality of the main image.

(39) An application scenario of the laminated glass **10** is introduced below. In the case where the laminated glass **10** is applied to a vehicle **1**, the laminated glass **10** is mounted to the vehicle **1** at a certain inclination angle to serve as a front windshield. The first transparent substrate **110** of the laminated glass **10** is a substrate exposed to the outside of the vehicle **1**. The second transparent substrate **120** is a substrate inside the vehicle **1**. To illustrate the beneficial effects in the case where

the laminated glass **10** includes the masking layer **140**, a case where the laminated glass **10** does not include the masking layer **140** will be described first. The projection device in the vehicle may project the first image P**1** onto the laminated glass **10** to form the first image P**1** on the second transparent substrate **120**. Objects outside the vehicle can also be seen from the inside of the vehicle through the laminated glass **10**. The projection device in the vehicle can emit a light ray to project the first image P**1** onto the laminated glass **10**, where the light ray is incident on the fourth surface **122** to enter the laminated glass **10**, where the light ray can be reflected by the fourth surface **122** to form the reflected light B and be reflected by the first surface **111** to form the reflected light A. The reflected light B and the reflected light A do not coincide to generate a reflection ghost. A light ray from an object outside the vehicle can be incident on the first surface **111**, enter the laminated glass **10**, and then pass through the laminated glass **10** to enter the vehicle to form incident light C. The incident light C may form transmission ghosting due to the inclined mounting and the parallel thickness of the laminated glass **10**. The laminated glass **10** in the embodiments of the disclosure includes the masking layer **140**, which can reduce or even block the reflected light A and the incident light C, thereby weakening or even blocking the reflection ghosting and the transmission ghosting.

(40) In conclusion, the laminated glass **10** provided in the embodiments of the disclosure has the masking layer **140** located in the light-blocking region R**10**, which can reduce or even block the reflection ghosting and the transmission ghosting. It can be seen that the laminated glass **10** provided in the embodiments of the disclosure can enhance the quality of the image projected thereon.

(41) Referring to FIG. **1** again, the light-blocking region R**10** has the first region R**110** located blow the light-transmitting region R**20**. The first region R**110** has one or more first function display regions R**111** for displaying of the first image P**1**. The light-blocking region R**10** further has a second region R**120** and a third region R**130**. The second region R**120** is located above the light-transmitting region R**20**. The third region R**130** is located at two sides of the light-transmitting region R**20**. Both the second region R**120** and the third region R**130** are configured for shielding electronic components or wiring.

(42) It may be noted that the light-blocking region R**10** is disposed to surround the light-transmitting region R**20**. That is, the first region R**110**, the second region R**120**, and the third region R**130** are located in the light-blocking region R**10** and cooperatively surround the light-transmitting region R**20**.

(43) In the embodiments, the light-blocking region R**10** is divided into three regions. The first region R**110** has one or more first function display regions R**111**. In the case where the first region R**110** has multiple first function display regions R**111**, the multiple first function display regions R**111** may be arranged separately or formed integrally. Alternatively, some of the multiple first function display regions R**111** are arranged separately and the rest of the multiple first function display regions R**111** are formed integrally. Each of the first function display regions R**111** is used for displaying of a corresponding one first image P**1**. Optionally, a ratio of a total area of the one or more first function display region R**111** to the area of the first region R**110** is greater than 10%, achieving better display effects of the first image P**1**. The second region R**120** and the third region R**130** on the one hand serve for shielding electronics or wiring mounted in later applications.

(44) Referring to FIG. **4** again, in an embodiment, the first function display region R**111** is part of the fourth surface **122**, a proportion of S-polarized light in the projection light ray incident ranges from 60% to 100%, and the first function display region R**111** has the reflectivity for the projection light ray incident greater than or equal to 8%.

(45) In the embodiments, preferably, the proportion of S-polarized light in the projection light ray is 100%, so that the reflectivity of the first function display region R**111** for the projection light ray incident can be further improved, thereby improving the clarity of the first image P**1**.

(46) Referring to FIGS. **5**, **6**, and **7** together, FIG. **5** is an exploded cross-sectional structural view

of the laminated glass **10** taken along line I-I of FIG. **1** according to another embodiment of the disclosure, FIG. **6** is an exploded cross-sectional structural view of the laminated glass **10** taken along line I-I of FIG. **1** according to another embodiment of the disclosure, and FIG. **7** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **1** according to another embodiment of the disclosure. The adhesive film **130** is of uniform thickness. The laminated glass **10** further includes a dielectric film **150**. The dielectric film **150** may be disposed on the third surface **121** (see FIG. **6**). Alternatively, the dielectric film **150** may be disposed on the fourth surface **122** (see FIG. **5**). Alternatively, the dielectric film **150** may be wrapped in the adhesive film **130** (see FIG. **7**). The dielectric film **150** is located in the first region R**110**. An orthographic projection of the dielectric film **150** on the second transparent substrate **120** covers all of the first function display regions R**111**. The dielectric film **150** can reflect S-polarized light.

(47) In the embodiments, the dielectric film **150** can reflect S-polarized light. The dielectric film **150** may be disposed on the fourth surface **122** (see FIG. **5**). Alternatively, the dielectric film **150** may be disposed on the third surface **121** (see FIG. **6**). Alternatively, the dielectric film **150** may be wrapped in the adhesive film **130** (see FIG. **7**). A combination of the second transparent substrate **120** and the dielectric film **150** allows the second transparent substrate **120** to reflect S-polarized light. For example, in the case where a proportion of S-polarized light in the projection light ray incident on one side of the second transparent substrate **120** is relatively large (for example, ranges from 60% to 100%), the second transparent substrate **120** has a relatively high reflectivity for the projection light ray incident on one side of the second transparent substrate **120** in the light-blocking region R**10**. For example, the second transparent substrate **120** has a reflectivity of 22% for the projection light ray incident at an angle of 60° in the light-blocking region R**10**. Preferably, a proportion of S-polarized light in the projection light ray is 100%, thereby further weakening or even eliminating light reflected by the first transparent substrate **110**. The orthographic projection of the dielectric film **150** on the second transparent substrate **120** covers all of the first function display regions R**111**, thereby further improving the brightness and clarity of reflection of the second transparent substrate **120** for the light incident on one side of the second transparent substrate **120**.

(48) Referring to FIGS. **5**, **6**, and **7**, FIG. **5** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **1** according to another embodiment of the disclosure, FIG. **6** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **1** according to another embodiment of the disclosure, and FIG. **7** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **1** according to another embodiment of the disclosure. The adhesive film **130** is of uniform thickness. The laminated glass **10** further includes the dielectric film **150**. The dielectric film **150** may be disposed on the third surface **121** (see FIG. **6**). Alternatively, the dielectric film **150** may be disposed on the fourth surface **122** (see FIG. **5**). Alternatively, the dielectric film **150** may be wrapped in the adhesive film **130** (see FIG. **7**). The dielectric film **150** is located in the first region R**110**. The orthographic projection of the dielectric film **150** on the second transparent substrate **120** covers all of the first function display regions R**111**. The dielectric film **150** can reflect P-polarized light.

(49) In the embodiments, the dielectric film **150** can reflect P-polarized light. The dielectric film **150** may be, but is not limited to, a high refractive index layer, a low refractive index layer, a metal film (including one to five silver layers), or a laminated polyethylene terephthalate (PET). The dielectric film **150** may be disposed on the fourth surface **122** (see FIG. **5**). Alternatively, the dielectric film **150** may be disposed on the third surface **121** (see FIG. **6**). Alternatively, the dielectric film **150** may be wrapped in the adhesive film **130** (see FIG. **7**). A combination of the second transparent substrate **120** and the dielectric film **150** allows the second transparent substrate **120** to reflect P-polarized light. For example, in the case where a proportion of P-polarized light in the projection light ray incident on one side of the second transparent substrate **120** is relatively

large (for example, ranges from 60% to 100%), the second transparent substrate **120** has a reflectivity for the P-polarized light in the projection light ray incident on one side of the second transparent substrate **120** in the light-blocking region R**10**. For example, the second transparent substrate **120** has a reflectivity of 20% for the projection light ray incident at an angle of 65° in the light-blocking region R**10**. Preferably, a proportion of P-polarized light in the projection light ray is 100%, thereby further weakening or even eliminating light reflected by the first transparent substrate **110**, and improving the brightness and clarity of reflection of the second transparent substrate **120** for the light incident on one side of the second transparent substrate **120**. It is also possible for the driver wearing sunglasses to view the first image P**1** in the first function display region R**111**.

(50) Referring to FIGS. **8** and **9**, FIG. **8** is a schematic structural view illustrating region division of the laminated glass provided in another embodiment of the disclosure, and FIG. **9** is a schematic structural view illustrating region division of the laminated glass provided in another embodiment of the disclosure. The laminated glass **10** further includes one or more flexible display screens **160**. The flexible display screens **160** are located in the first region R**110**. Each of the one or more flexible display screens **160** is disposed in a corresponding one of the first function display regions R**111**. The flexible display screen **160** is configured to display the first image P**1**. Alternatively, the laminated glass **10** further includes one or more first projection light sources **170**. The first projection light source **170** is configured to project the first image P**1** onto the first function display region R**111**. Each of the first projection light sources **170** is disposed corresponding to one of the first function display regions R**111**.

(51) Referring to FIG. **8**, in the embodiments, the flexible display screens **160** are disposed in the first function display regions R**111**. Each of the flexible display screens **160** corresponds to one of the first function display regions R**111**. Each of the flexible display screens **160** may be disposed between the masking layer **140** and the third surface **121**. Alternatively, each of the flexible display screens **160** may be disposed on the fourth surface **122**. The flexible display screen **160** may be, but not limited to, a mini light-emitting diode (LED) display screen, a micro LED display screen, or an organic light-emitting diode (OLED) display screen. The flexible display screen **160** can generate an image directly. Light rays emitted by the flexible display screen **160**, for forming the first image P**1**, can directly pass through the second transparent substrate **120** or do not need to pass through the second transparent substrate **120**, without being affected by reflection of the first transparent substrate **110**, thereby further avoiding ghosting caused by reflection of the first transparent substrate **110** for the light rays emitted and reflection of the second transparent substrate **120** for the light rays emitted.

(52) In another embodiment, referring to FIG. **9**, each first projection light source **170** is disposed corresponding to one first function display region R**111**. The first projection light source **170** is disposed at one side of the second transparent substrate **120**. Optionally, a proportion of S-polarized light in light rays emitted by the first projection light source **170** ranges from 60% to 100%. With the aid of the dielectric film **150** that can reflect S-polarized light, the clarity of the first image P**1** can be improved. Preferably, the proportion of S-polarized light in the light rays emitted by the first projection light source **170** is 100%, so that the clarity of the first image P**1** can be further improved.

(53) In yet another embodiment, in the case where the laminated glass **10** has multiple first function display regions R**111**, the flexible display screens **160** are used in combination with the first projection light source **170**. The flexible display screens **160** are arranged corresponding to some of the first function display regions R**111**, and the first projection light sources **170** are arranged corresponding to the rest of the first function display regions R**111**. In the embodiments, it can not only weaken ghosting caused by reflection of the first transparent substrate **110** for the light rays emitted and reflection of the second transparent substrate **120** for the light rays emitted, but also increase the display diversity of the first function display regions R**111**. As such, mounting of the

laminated glass **10** can be optimized according to practical applications.

(54) It may be noted that the flexible display screen **160** or the first function display region R**111** is closer to the fourth surface **122** than the masking layer **140**, so that the masking layer **140** can serve as the display background of the first image P**1**. The masking layer **140** may be, but not limited to, a dark ink layer or a colored polymer film. Here, a projection display distance for the first image P**1** ranges from 0.5 m to 5 m. In some embodiments, a projection display distance for the first image P**1** is a distance between the first projection light source **170** and the first image P**1**.

(55) Referring to FIG. **10**, FIG. **10** is a schematic structural view illustrating region division of the laminated glass provided in another embodiment of the disclosure. The light-transmitting region R**20** further includes a primary viewing region R**210**. A lower boundary of the primary viewing region R**210** is at least 25 mm higher than an upper boundary of the first region R**110**.

(56) In the embodiments, the lower boundary of the primary viewing region R**210** is at least 25 mm higher than the upper boundary of the first region R**110**, so that an optical sensitive zone can be avoided, and optical distortion between the first region R**110** and the primary viewing region R**210** can be avoided, where the optical distortion may interfere with an imaging in the first region R**110** and an imaging in the primary viewing region R**210**.

(57) Referring to FIG. **10** again, the primary viewing region R**210** further has one or more second function display regions R**211**. The second function display region R**211** is configured to display a second image P**2**. A projection display distance for the second image P**2** is greater than or equal to 7.5 m. In some embodiments, a projection display distance for the second image P**2** is a distance between a second projection light source **190** and the second image P**2**.

(58) In the embodiments, the laminated glass **10** further has the second function display region R**211**. An area of the second function display region R**211** is larger than that of the first function display region R**111**, so that the laminated glass **10** can display the second image P**2** larger than the first image P**1**, thereby enriching the image display of the laminated glass **10**. A projection light ray for forming the first image P**1** is incident on the first function display region R**111** at an angle of 50° to 72°, and the first function display region has a reflectivity, for the projection light ray for forming the first image P**1**, greater than or equal to 4%. A projection light ray for forming the second image P**2** is incident on the second function display region R**211** at an angle of 50° to 72°, and the second function display region R**211** has a reflectivity, for the projection light ray for forming the second image P**2**, greater than or equal to 8%.

(59) Referring to FIGS. **11** and **12**, FIG. **11** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure, and FIG. **12** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure. A thickness of the adhesive film **130** gradually decreases in a direction from the second region R**120** to the first region R**110**. An orthographic projection of the adhesive film **130** on the second transparent substrate **120** covers all of the second function display regions R**211**.

(60) In the embodiments, the thickness of the adhesive film **130** gradually decreases in the direction from the second region R**120** to the first region R**110**. In other words, the adhesive film **130** is wedge-shaped. Optionally, a wedge-shaped angle caused by a gradual vary in thickness of the adhesive film **130** ranges from 0.15 mrad to 0.55 mrad. An orthographic projection of part of the adhesive film **130**, which gradually varies in thickness, on the second transparent substrate **120** covers at least all of the second function display regions R**211**. The second function display region R**211** servers as part of the fourth surface **122**. A proportion of S-polarized light in the projection light ray incident for forming the second image P**2** ranges from 60% to 100%. The second function display region R**211** has a reflectivity greater than or equal to 8% for the projection light ray, for forming the second image P**2**, incident at an angle of 50° to 72°. Preferably, the proportion of S-polarized light in the projection light ray for forming the second image P**2** is 100%. The adhesive film **130** that gradually varies in thickness can correct, by means of a wedge-shaped angle, ghosting

in the second function display region R**211** caused by reflection of the first transparent substrate **110** for the projection light ray incident and refection of the second transparent substrate **120** for the projection light ray incident. In addition, in the case where the laminated glass **10** has multiple second function display regions R**211**, wedge-shaped angles of the adhesive film **130** in different second function display regions R**211** may be either equal or unequal. Since the second function display regions R**211** differ in size, shape, and position, and incident angles of light sources are also different, different wedge-shaped angles are required to correct ghosting in each second function display region R**211** caused by reflection of the first transparent substrate **110** for the projection light ray and reflection of the second transparent substrate **120** for the projection light ray. In the case where each of the second function display regions R**211** has consistent setting conditions, the same wedge-shaped may also be used.

(61) Referring to FIG. **11**, in an embodiment, the masking layer **140** is disposed on the third surface **121**. The adhesive film **130** is disposed between the masking layer **140** and the second surface **112**. An orthographic projection of part of the adhesive film **130**, which gradually varies in thickness, on the second transparent substrate **120** covers all of the second function display regions R**211**. The adhesive film **130** may correct ghosting in the second function display region R**211** caused by reflection of the first transparent substrate **110** for the projection light ray and reflection of the second transparent substrate **120** for the projection light ray, thereby improving the clarity of the second image P**2**. In another embodiment, referring to FIG. **12**, the masking layer **140** is disposed on the second surface **112**. The adhesive film **130** is disposed between the masking layer **140** and the third surface **121**. An orthographic projection of part of the adhesive film **130**, which gradually varies in thickness, on the second transparent substrate **120** covers all of the second function display regions R**211**. The adhesive film **130** may correct ghosting in the second function display region R**211** caused by the reflection of the first transparent substrate **110** for the projection light ray and reflection of the second transparent substrate **120** for the projection light ray, thereby improving the clarity of the second image P**2**. Alternatively, an orthographic projection of part of the adhesive film **130**, which gradually varies in thickness, on the second transparent substrate **120** covers all of the first function display regions R**111** and all of the second function display regions R**211**. The adhesive film **130** may correct ghosting in the first function display region R**111**, caused by reflection of the first transparent substrate **110** for the projection light ray and reflection of the second transparent substrate **120** for the projection light ray, and ghosting in the second function display region R**211**, caused by reflection of the first transparent substrate **110** for the projection light ray and reflection of the second transparent substrate **120** for the projection light ray, thereby not only improving the clarity of the second image P**2**, but also further improving the clarity of the first image P**1**. Further, the manufacturing efficiency of the adhesive film **130**, the first transparent substrate **110**, and the second transparent substrate **120** can be improved.

(62) Referring to FIGS. **13** to **16**, FIG. **13** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure, FIG. **14** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure, FIG. **15** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure, and FIG. **16** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure. The laminated glass **10** further includes the dielectric film **150** disposed on the third surface **121** or the fourth surface **122**. The dielectric film **150** can reflect P-polarized light or S-polarized light. Alternatively, the dielectric film **150** is disposed on the fourth surface **122**, and the dielectric film **150** has anti-reflection capability for S-polarized light and a reflectivity for S-polarized light less than 6%. Alternatively, the dielectric film **150** is a laminated structure formed by a high refractive index layer and a low refractive index layer and disposed on the third surface **121** or the fourth surface **122**, and can reflect P-polarized light or S-polarized light. Alternatively, the dielectric film

150 includes at least one metal layer (one to five silver layers) and is disposed on the second surface **112** or the third surface **121**, and can reflect P-polarized light. Alternatively, the dielectric film **150** is a laminated PET and sandwiched between the second surface **112** and the third surface **121**, and can reflect P-polarized light. The orthographic projection of the dielectric film **150** on the second transparent substrate **120** covers at least all of the second function display regions R**211**. In an embodiment, the orthographic projection of the dielectric film **150** on the second transparent substrate **120** cover all of the second function display regions R**211**. In another embodiment, the orthographic projection of the dielectric film **150** on the second transparent substrate **120** covers all of the second function display regions R**211** and all of the first function display regions R**111**.
(63) Referring to FIG. **13**, on the basis of the embodiment illustrated in FIG. **11**, the dielectric film **150** is disposed on the fourth surface **122**. In an embodiment, in the case where a proportion of P-polarized light in a light ray emitted by a light source at one side of the fourth surface **122** ranges from 60% to 100%, the dielectric film **150** can reflect P-polarized light and have a reflectivity for P-polarized light greater than 10%. The first surface **111** of the first transparent substrate **110** has a very low reflectivity for P-polarized light incident at an angle of 50° to 72°. For example, the first surface **111** has a reflectivity for P-polarized light incident at an angle of 57° less than 1%, so that reflection of the first surface **111** for the light ray emitted can be weakened, that is, ghosting caused by reflection of the first surface **111** and reflection of the fourth surface **122**, for the light ray emitted by the light source at one side of the fourth surface **122**, can be further weakened. In another embodiment, in the case where a proportion of S-polarized light in a light ray emitted by the light source at one side of the fourth surface **122** ranges from 60% to 100%, the dielectric film **150** can reflect S-polarized light and have a reflectivity for S-polarized light not less than 28%, thereby weakening reflection of the first surface **111** for the light ray emitted by the light source. In addition, with the aid of the adhesive film **130** that varies in thicknesses, image superposing between reflection of the first surface **111** and reflection of the fourth surface **122**, for the light ray emitted by the light source at one side of the fourth surface **122**, can be strengthened, that is, ghosting caused by reflection of the first surface **111** and reflection of the fourth surface **122**, for the light ray emitted by the light source at one side of the fourth surface **122**, can be weakened. In a further embodiment, in the case where a proportion of S-polarized light in the light ray emitted by the light source at one side of the fourth surface **122** ranges from 60% to 100%, the orthographic projection of the dielectric film **150** on the second transparent substrate **120** does not cover all of the first function display regions R**111**, the dielectric film **150** has anti-reflection capability for S-polarized light and a reflectivity for S-polarized light of not greater than 6%, so that reflection of the fourth surface **122** for the light ray emitted by the light source at one side of the fourth surface **122** can be weakened, thereby weakening ghosting caused by reflection of the first surface **111** and reflection of the fourth surface **122** for the light ray emitted by the light source at one side of the fourth surface **122**.
(64) Referring to FIG. **14**, on the basis of the embodiment illustrated in FIG. **12**, the dielectric film **150** is disposed on the fourth surface **122**. In an embodiment, in the case where a proportion of P-polarized light in a light ray emitted by a light source at one side of the fourth surface **122** ranges from 60% to 100%, the dielectric film **150** can reflect P-polarized light and have a reflectivity for P-polarized light greater than 10%. The first surface **111** of the first transparent substrate **110** has a very low reflectivity for P-polarized light incident at an angle of 50° to 72°. For example, the first surface **111** has a reflectivity for P-polarized light incident at an angle of 57° less than 1%, so that reflection of the first surface **111** for the light ray emitted can be weakened, that is, ghosting caused by reflection of the first surface **111** and reflection of the fourth surface **122**, for a light ray emitted by the light source at one side of the fourth surface **122**, can be weakened. In another embodiment, in the case where a proportion of S-polarized light in a light ray emitted by the light source at one side of the fourth surface **122** ranges from 60% to 100%, the dielectric film **150** can reflect S-polarized light and have a reflectivity for S-polarized light not less than 28%, thereby weakening

reflection of the first surface **111** for the light ray emitted by the light source. In addition, with the aid of the adhesive film **130** that varies in thicknesses, image superposing between reflection of the first surface **111** and reflection of the fourth surface **122**, for the light ray emitted by the light source at one side of the fourth surface **122**, can be strengthened, that is, ghosting caused by reflection of the first surface **111** and reflection of the fourth surface **122**, for the light ray emitted by the light source at one side of the fourth surface **122**, can be weakened. In a further embodiment, in the case where a proportion of S-polarized light in the light ray emitted by the light source at one side of the fourth surface **122** ranges from 60% to 100%, the orthographic projection of the dielectric film **150** on the second transparent substrate **120** does not cover all of the first function display regions R**111**, the dielectric film **150** has anti-reflection capability for S-polarized light and a reflectivity for S-polarized light of not greater than 6%, so that reflection of the fourth surface **122** for the light ray emitted by the light source at one side of the fourth surface **122** can be weakened, thereby weakening ghosting caused by reflection of the first surface **111** and reflection of the fourth surface **122** for the light ray emitted by the light source at one side of the fourth surface **122**.

(65) Referring to FIG. **15**, on the basis of the embodiment illustrated in FIG. **11**, the dielectric film **150** is disposed between the masking layer **140** and the adhesive film **130**. The orthographic projection of the dielectric film **150** on the second transparent substrate **120** covers all of the second function display regions R**211**. In an embodiment, in the case where a proportion of P-polarized light in a light ray emitted by a light source at one side of the fourth surface **122** ranges from 60% to 100%, the dielectric film **150** can reflect P-polarized light and have a reflectivity for P-polarized light greater than 10%. Each of the first surface **111** of the first transparent substrate **110** and the fourth surface **122** of the second transparent substrate **120** has a very low reflectivity for P-polarized light incident at an angle of 50° to 72°. For example, each of the first surface **111** and the fourth surface **122** has a reflectivity for P-polarized light incident at an angle of 57° less than 1%, so that reflection of the first surface **111** for the light ray emitted and the reflection of the fourth surface **111** for the light ray emitted can be weakened, that is, ghosting caused by reflection of the first surface **111** and reflection of the fourth surface **122**, for the light ray emitted by the light source at one side of the fourth surface **122**, can be weakened.

(66) Referring to FIG. **16**, on the basis of the embodiment illustrated in FIG. **12**, the dielectric film **150** is disposed between the adhesive film **130** and the third surface **121**. In an embodiment, in the case where a proportion of P-polarized light in a light ray emitted by a light source at one side of the fourth surface **122** ranges from 60% to 100%, the dielectric film **150** can reflect P-polarized light and have a reflectivity for P-polarized light greater than 10%, so that reflection of the first surface **111** for the light ray emitted can be weakened, that is, ghosting caused by reflection of the first surface **111** and reflection of the fourth surface **122**, for the light ray emitted by the light source at one side of the fourth surface **122**, can be weakened.

(67) Referring to FIGS. **17** to **22**, FIG. **17** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure, FIG. **18** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure, FIG. **19** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure, FIG. **20** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure, FIG. **21** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure, and FIG. **22** is an exploded cross-sectional structural view of the laminated glass taken along line I-I of FIG. **10** according to another embodiment of the disclosure. The adhesive film **130** is of uniform thickness. The laminated glass **10** further includes the dielectric film **150**. The dielectric film **150** may be disposed on the third surface **121**. Alternatively, the dielectric film **150** may be disposed on the fourth surface **122**.

Alternatively, the dielectric film **150** may be wrapped in the adhesive film **130**. The dielectric film **150** can reflect P-polarized light. Alternatively, the dielectric film **150** can reflect polarized light and have a reflectivity for polarized light less than 6%. An orthographic projection of the dielectric film **150** on the second transparent substrate **120** covers at least all of the second function display regions R**211**.

(68) Referring to FIG. **17**, on the basis of the embodiment illustrated in FIG. **4**, the dielectric film **150** is disposed on the fourth surface **122**. In an embodiment, in the case where a proportion of P-polarized light in a light ray emitted by a light source at one side of the fourth surface **122** ranges from 60% to 100%, the dielectric film **150** can reflect P-polarized light and have a reflectivity for P-polarized light greater than 10%. The first surface **111** of the first transparent substrate **110** has a very low reflectivity for P-polarized light incident at an angle of 50° to 72°. For example, the first surface **111** has a reflectivity for P-polarized light incident at an angle of 57° less than 1%, so that reflection of the first surface **111** for the light ray emitted can be weakened, that is, ghosting caused by reflection of the first surface **111** and reflection of the fourth surface **122**, for the light ray emitted by the light source at one side of the fourth surface **122**, can be weakened. In another embodiment, in the case where a proportion of S-polarized light in the light ray emitted by the light source at one side of the fourth surface **122** ranges from 60% to 100%, the orthographic projection of the dielectric film **150** on the second transparent substrate **120** only covers all of the first function display regions R**111**, the dielectric film **150** has anti-reflection capability for S-polarized light and a reflectivity for S-polarized light of not greater than 6%, so that reflection of the fourth surface **122** for the light ray emitted by the light source at one side of the fourth surface **122** can be weakened, thereby weakening ghosting caused by reflection of the first surface **111** and reflection of the fourth surface **122** for the light ray emitted by the light source at one side of the fourth surface **122**.

(69) Referring to FIG. **18**, on the basis of the embodiment illustrated in FIG. **3**, the dielectric film **150** is disposed on the fourth surface **122**. In an embodiment, in the case where a proportion of P-polarized light in a light ray emitted by a light source at one side of the fourth surface **122** ranges from 60% to 100%, the dielectric film **150** can reflect P-polarized light and have a reflectivity for P-polarized light greater than 10%. The first surface **111** of the first transparent substrate **110** has a very low reflectivity for P-polarized light incident at an angle of 50° to 72°. For example, the first surface **111** has a reflectivity for P-polarized light incident at an angle of 57° less than 1%, so that reflection of the first surface **111** for the light ray emitted can be weakened, that is, ghosting caused by reflection of the first surface **111** and reflection of the fourth surface **122**, for a light ray emitted by the light source at one side of the fourth surface **122**, can be weakened. In another embodiment, in the case where a proportion of S-polarized light in the light ray emitted by the light source at one side of the fourth surface **122** ranges from 60% to 100%, the orthographic projection of the dielectric film **150** on the second transparent substrate **120** does not cover all of the first function display regions R**111**, the dielectric film **150** has anti-reflection capability for S-polarized light and a reflectivity for S-polarized light of not greater than 6%, so that reflection of the fourth surface **122** for the light ray emitted by the light source at one side of the fourth surface **122** can be weakened, thereby weakening ghosting caused by reflection of the first surface **111** and reflection of the fourth surface **122** for the light ray emitted by the light source at one side of the fourth surface **122**.

(70) Referring to FIG. **19**, on the basis of the embodiment illustrated in FIG. **4**, the dielectric film **150** is disposed between the adhesive film **130** and the masking layer **140**. In an embodiment, in the case where a proportion of P-polarized light in a light ray emitted by a light source at one side of the fourth surface **122** ranges from 60% to 100%, the dielectric film **150** can reflect P-polarized light and have a reflectivity for P-polarized light greater than 10%. Each of the first surface **111** of the first transparent substrate **110** and the fourth surface **122** of the second transparent substrate **120** has a very low reflectivity for P-polarized light incident at an angle of 50° to 72°. For example,

each of the first surface **111** and the fourth surface **122** has a reflectivity for P-polarized light incident at an angle of 57° less than 1%, so that reflection of the first surface **111** for the light ray emitted and the reflection of the fourth surface **111** for the light ray emitted can be weakened, that is, ghosting caused by reflection of the first surface **111** and reflection of the fourth surface **122**, for the light ray emitted by the light source at one side of the fourth surface **122**, can be weakened.
(71) Referring to FIG. **20**, on the basis of the embodiment illustrated in FIG. **3**, the dielectric film **150** is disposed between the adhesive film **130** and the third surface **121**. In an embodiment, in the case where a proportion of P-polarized light in a light ray emitted by a light source at one side of the fourth surface **122** ranges from 60% to 100%, the dielectric film **150** can reflect P-polarized light and have a reflectivity for P-polarized light greater than 10%. Each of the first surface **111** of the first transparent substrate **110** and the fourth surface **122** of the second transparent substrate **120** has a very low reflectivity for P-polarized light incident at an angle of 50° to 72°. For example, each of the first surface **111** and the fourth surface **122** has a reflectivity for P-polarized light incident at an angle of 57° less than 1%, so that reflection of the first surface **111** for the light ray emitted and the reflection of the fourth surface **111** for the light ray emitted can be weakened, that is, ghosting caused by reflection of the first surface **111** and reflection of the fourth surface **122**, for the light ray emitted by the light source at one side of the fourth surface **122**, can be weakened.
(72) Referring to FIG. **21**, on the basis of the embodiment illustrated in FIG. **4**, the dielectric film **150** is disposed in the adhesive film **130**, so that the dielectric film **150** can be wrapped in the adhesive film **130**. In an embodiment, in the case where a proportion of P-polarized light in a light ray emitted by a light source at one side of the fourth surface **122** ranges from 60% to 100%, the dielectric film **150** can reflect P-polarized light and have a reflectivity for P-polarized light greater than 10%. Each of the first surface **111** of the first transparent substrate **110** and the fourth surface **122** of the second transparent substrate **120** has a very low reflectivity for P-polarized light incident at an angle of 50° to 72°. For example, each of the first surface **111** and the fourth surface **122** has a reflectivity for P-polarized light incident at an angle of 57° less than 1%, so that reflection of the first surface **111** for the light ray emitted and the reflection of the fourth surface **111** for the light ray emitted can be weakened, that is, ghosting caused by reflection of the first surface **111** and reflection of the fourth surface **122**, for the light ray emitted by the light source at one side of the fourth surface **122**, can be weakened.
(73) Referring to FIG. **22**, on the basis of the embodiment illustrated in FIG. **3**, the dielectric film **150** is disposed in the adhesive film **130**, so that the dielectric film **150** can be wrapped in the adhesive film **130**. In an embodiment, in the case where a proportion of P-polarized light in a light ray emitted by a light source at one side of the fourth surface **122** ranges from 60% to 100%, the dielectric film **150** can reflect P-polarized light and have a reflectivity for
(74) P-polarized light greater than 10%. Each of the first surface **111** of the first transparent substrate **110** and the fourth surface **122** of the second transparent substrate **120** has a very low reflectivity for P-polarized light incident at an angle of 50° to 72°. For example, each of the first surface **111** and the fourth surface **122** has a reflectivity for P-polarized light incident at an angle of 57° less than 1%, so that reflection of the first surface **111** for the light ray emitted and the reflection of the fourth surface **111** for the light ray emitted can be weakened, that is, ghosting caused by reflection of the first surface **111** and reflection of the fourth surface **122**, for the light ray emitted by the light source at one side of the fourth surface **122**, can be weakened.
(75) Referring to FIGS. **23** and **24**, FIG. **23** is a schematic structural view illustrating region division of the laminated glass **10** provided in another embodiment of the disclosure, and FIG. **24** is an exploded cross-sectional structural view of the laminated glass **10** taken along line I-I of FIG. **23** according to another embodiment of the disclosure. The laminated glass **10** further includes a colored region R**30**. The colored region R**30** is located at one side of the light-blocking region R**10** away from the light-transmitting region R**20**. The laminated glass **10** further includes a colored layer **180** carried on the second transparent substrate **120**. The colored layer **180** is located in the

colored region R**30**. The colored layer **180** is used for alignment and vehicle window fixing during mounting of the laminated glass **10**, or disposed on a surface of an adhesive substrate of a fixing member.
(76) In the embodiments, the colored layer **180** is disposed on an outermost surface of the laminated glass **10** and at one side of the fourth surface **122**. An orthographic projection of the colored layer **180** on the second transparent substrate **120** covers the colored region R**30**. The colored region R**30** can be configured to shield electronic components or wiring mounted later, and can also be configured to assist in mounting of the laminated glass **10** onto other devices. For example, the colored region R**30** facilitates gluing, aligning, or enhancement in adhesive strength. Furthermore, the upper boundary of the first region R**110** is higher than an upper boundary of the colored region R**30** in the first region R**110**. Optionally, the upper boundary of the first region R**110** is at least 80 mm higher than the upper boundary of the colored region R**30** in the first region R**110**, so that sufficient space is reserved for the first function display region R**111**.
(77) Referring to FIG. **25**, FIG. **25** is a schematic structural view illustrating region division of the laminated glass provided in another embodiment of the disclosure. The laminated glass **10** further includes one or more first projection light sources **170** and one or more second projection light sources **190**. The first projection light source **170** is configured to project the first image P**1** to the first function display region R**111**. Each of the first projection light sources **170** corresponds to one of the first function display regions R**111**. The second projection light source **190** is configured to project the second image P**2** to the second function display region R**211**. Each of the second projection light sources **190** corresponds to one of the second function display regions R**211**.
(78) In the embodiments, the second projection light source **190** is configured to project an image to the second function display region R**211**, so that the second image P**2** which has a larger size can be presented, thereby increasing the display diversity of the laminated glass **10**.
(79) Referring to FIG. **26**, FIG. **26** is a schematic structural view illustrating region division of the laminated glass provided in another embodiment of the disclosure. The laminated glass **10** further includes one or more flexible display screens **160** and one or more second projection light sources **190**. The flexible display screens **160** are arranged in the first region R**110**. Each of the flexible display screens **160** is disposed corresponding to one of the first function display regions R**111**. The flexible display screen **160** is configured to display the first image P**1**. The second projection light source **190** is configured to project the second image P**2** onto the second function display region R**211**. Each of the one or more second projection light sources **190** is disposed corresponding to one of the second function display regions R**211**.
(80) In the embodiments, the second projection light source **190** is configured to project image onto the second function display region R**211**, so that the second image P**2** which has a larger size can be presented, thereby increasing the display diversity of the laminated glass **10**.
(81) A head-up display system is provided in the disclosure. In an embodiment (as illustrated in FIG. **9**), the head-up display system includes the first projection light source **170** and the laminated glass **10** described in any one of embodiments where only the first function display region R**111** is provided. In another embodiment (as illustrated in FIG. **25**), the head-up system includes the first projection light source **170**, the second projection light source **190**, and the laminated glass **10** described in any one of embodiments where the second function display region R**211** is provided.
(82) In an embodiment, a proportion of P-polarized light in the projection light ray for forming the first image P**1** ranges from 60% to 100%, and a proportion of S-polarized light in the projection light ray for forming the second image P**2** ranges from 60% to 100%.
(83) In another embodiment, a proportion of S-polarized light in the projection light ray for forming the first image P**1** ranges from 60% to 100%, and a proportion of P-polarized light in the projection light ray for forming the second image P**2** ranges from 60% to 100%.
(84) In a further embodiment, a proportion of P-polarized light in the projection light ray for forming the first image P**1** ranges from 60% to 100%, and a proportion of P-polarized light in the

projection light ray for forming the second image P**2** ranges from 60% to 100%.
(85) In a further embodiment, a proportion of S-polarized light in the projection light ray for forming the first image P**1** ranges from 60% to 100%, and a proportion of S-polarized light in the projection light ray for forming the second image P**2** ranges from 60% to 100%.
(86) It may be noted that, preferably, a proportion of S-polarized light in the projection light ray is 100%, or proportion of P-polarized light in the projection light ray is 100%, thereby achieving a better projection effect.
(87) Referring to FIG. **27**, FIG. **27** is a schematic view of a vehicle provided the disclosure. The vehicle **1** is further provided in the disclosure. The vehicle **1** includes the laminated glass **10** described in any one of the above embodiments. The vehicle **1** further includes a vehicle body **20**. The laminated glass **10** is disposed on the vehicle body **20**. For the laminated glass **10**, reference may be made to the foregoing illustrations, and details are not repeated herein. In the case where the laminated glass **10** is applied to the vehicle **1**, the first transparent substrate **110** is disposed on an outer side of the vehicle **1**, and the second transparent substrate **120** is disposed on an inner side of the vehicle **1**.
(88) In the embodiments, the vehicle **1** may be, but is not limited to, cars, multi-purpose automobiles (MPV), sports/suburban utility vehicles (SUV), off-road automobiles (ORV), pick-and-place automobiles, passenger cars, cargo cars, and the like. An included angle between the laminated glass **10** and the vertical plane is referred to as a mounting angle, and typically ranges from 50° to 72°. In the case where no masking layer **140** is provided in the laminated glass **10**, two instances of ghosting occur. On the one hand, projection light rays emitted from the interior of the vehicle **1** will be reflected by the first transparent substrate **110** and the second transparent substrate **120** to cause ghosting. On the other hand, ghosting may further be caused by light rays passing through the laminated glass **1** from objects outside the vehicle **1**, and reflected light formed by refection of the laminated glass **10** for the projection light rays emitted from the interior of the vehicle **1**. With the aid of the masking layer **140**, the above ghosting can be weakened or even eliminated. With the aid of the dielectric film **150**, the above ghosting can be further weakened or even eliminated, and ghosting in the second function region R**211** can be weakened or even eliminated. In the case where the mounting angle is 60°, experiments are conducted on reflection of the dielectric film **150**, with light-transmitting capabilities, for projection light rays in the first function display region R**111**. The obtained data is presented in the following two tables.
(89) TABLE-US-00001 TABLE 1 reflection data of the laminated glass without a transparent dielectric film for projection light rays in the first function display region of the light-blocking region Light Source Type Reflectivity Ordinary Light Source 7.5% P-polarized Light 0.3% S-polarized Light 13%
(90) TABLE-US-00002 TABLE 2 reflection data of the laminated glass with different transparent dielectric films for projection light rays in the first function display region Transparent dielectric film Type Light Source Type Reflectivity Anti-reflective film Ordinary Light Source 5.1% P-polarized Reflective P-polarized Light 11% film S-polarized Reflective S-polarized Light 22% film
(91) In Table 1, in the case where the light source type is an ordinary light source, light emitted by the ordinary light source is an irregular collection of countless polarized lights, so when directly observed, it is impossible to determine which direction the light intensity is biased towards. This type of light, where the intensity of the light waves vibrating in various directions is equal, is also known as natural light. In the case where the light source type is P-polarized light, a proportion of P-polarized light in light rays emitted by the light source ranges from 60% to 100%. In the case where the light source type is S-polarized light, a proportion of S-polarized light in light rays emitted by the light source ranges from 60% to 100%. The light source type in Table 2 can refer to the illustration of the light source type in Table 1, and will not be repeated herein. The anti-reflective film is the afore-mentioned dielectric film **150** which has anti-reflection capability for S-

polarized light and a relatively low reflectivity for S-polarized light (of not greater than 6%). The P-polarized reflective film is the afore-mentioned dielectric film **150** which can reflect P-polarized light. The S-polarized reflective film is the afore-mentioned dielectric film **150** which can reflect S-polarized light. The following can be seen from the experimental data in the above two tables. In an embodiment, in the first function display region R**111**, in the case where the first projection light source **170** is the ordinary light source and the anti-reflective film is mounted, a reflectivity of the fourth surface **122** for projection light rays emitted by the first projection light source **170** in the first function display region R**111** decreases from 7.5% to 5.1%. In another embodiment, in the first function display region R**111**, in the case where the first projection light source **170** is configured to emit P-polarized light and the P-polarized reflective film is mounted, a reflectivity of the fourth surface **122** for projection light rays emitted by the first projection light source **170** in the first function display region R**111** increases from 0.3% to 11%. In still another embodiment, in the first function display region R**111**, in the case where the first projection light source **170** is configured to emit S-polarized light and the S-polarized reflective film is mounted, the reflectivity of the fourth surface **122** for projection light rays emitted by the first projection light source **170** in the first function display region R**111** increases from 13% to 22%.
(92) Alternatively, the laminated glass **10** further has a transparent conductive layer. The transparent conductive layer is mounted between the first transparent substrate **110** and the second transparent substrate **120**. The transparent conductive layer has at least one of heat insulation capacity and heating function for reflecting infrared rays, and the transparent conductive layer covers at least 80% of the light-transmitting region R**20**.
(93) Optionally, the projection display distance for the first image P**1** ranges from 0.5 m to 5 m. The first image P**1** may indicate key information such as vehicle speed, fuel level, or the engine speed. Alternatively, the projection display distance of the second image P**2** is greater than or equal to 7.5 m. The second image P**2** may be a relatively large image which can indicate information such as route navigation, a speeding reminder, or obstacle warnings.
(94) Although the embodiments of the disclosure have been illustrated and described above, it can be understood that the above embodiments are exemplary and cannot be understood as limitations on the disclosure. Those skilled in the art can make changes, modifications, replacements, and variations for the above embodiments within the scope of the disclosure, and these improvements and modifications are also considered to fall into the protection scope of the disclosure.

Claims

1. A laminated glass, comprising: a first transparent substrate having a first surface and a second surface opposite the first surface; a second transparent substrate having a third surface and a fourth surface opposite the third surface, wherein the third surface is closer to the second surface than the fourth surface; and an adhesive film disposed between the second surface and the third surface and configured to adhere the first transparent substrate and the second transparent substrate; wherein the laminated glass has a light-transmitting region and a light-blocking region surrounding at least part of a periphery of the light-transmitting region; and wherein the light-transmitting region has a visible light transmittance greater than or equal to 70%, the light-blocking region has a visible light transmittance less than or equal to 5%, the light-blocking region has a first region located below the light-transmitting region, and the first region has one or more first function display regions for displaying of a first image.
2. The laminated glass of claim 1, further comprising at least one flexible display screen in the first function display region, and the at least one flexible display screen is disposed between the second surface and the third surface and comprises at least one of a mini light-emitting diode (LED) display screen, a micro LED display screen, or an organic light-emitting diode (OLED) display screen.

3. The laminated glass of claim 1, wherein the first function display region is configured to receive a projection light ray for forming the first image that is incident at an angle of 50° to 72°; and the first function display region has a reflectivity for the projection light ray incident greater than or equal to 4%.
4. The laminated glass of claim 3, wherein the first function display region is part of the fourth surface, a proportion of S-polarized light in the projection light ray incident ranges from 60% to 100%, and the first function display region has the reflectivity for the projection light ray incident greater than or equal to 8%.
5. The laminated glass of claim 3, further comprising a dielectric film disposed in the first function display region, the dielectric film being on the third surface or the fourth surface, wherein a proportion of P-polarized light in the projection light ray incident ranges from 60% to 100%, and the first function display region has the reflectivity for the projection light ray incident greater than or equal to 8%; or a proportion of S-polarized light in the projection light ray incident ranges from 60% to 100%, and the first function display region has the reflectivity for the projection light ray incident greater than or equal to 8%.
6. The laminated glass of claim 3, further comprising a metal film disposed in the first function display region, the metal film being on the third surface, wherein a proportion of P-polarized light in the projection light ray incident ranges from 60% to 100%, and the first function display region has the reflectivity for the projection light ray incident greater than or equal to 6%.
7. The laminated glass of claim 3, further comprising a laminated polyethylene terephthalate (PET) disposed in the first function display, wherein a proportion of P-polarized light in the projection light ray incident ranges from 60% to 100%, and the first function display region has the reflectivity for the projection light ray incident greater than or equal to 10%.
8. The laminated glass of claim 2, further comprising a dark ink layer or a colored polymer film disposed in the light-blocking region, wherein the dark ink layer is disposed on at least one of the second surface or the third surface, and the colored polymer film is disposed between the second surface and the third surface; and the flexible display screen is closer to the fourth surface than the dark ink layer or the colored polymer film.
9. The laminated glass of claim 3, further comprising a dark ink layer or a colored polymer film disposed in the light-blocking region, wherein the dark ink layer is disposed on at least one of the second surface or the third surface, and the colored polymer film is disposed between the second surface and the third surface; and the first function display region is closer to the fourth surface than the dark ink layer or the colored polymer film.
10. The laminated glass of claim 1, wherein the fourth surface has a colored region, an upper boundary of the first region is at least 80 mm higher than an upper boundary of the colored region located in the first region.
11. The laminated glass of claim 1, wherein the light-transmitting region has one or more second function display regions, and the one or more second function display regions are configured for displaying of a second image.
12. The laminated glass of claim 11, wherein a projection display distance for the first image ranges from 0.5 m to 5 m, and a projection display distance for the second image is greater than or equal to 7.5 m.
13. The laminated glass of claim 12, wherein a projection light ray for forming the first image is incident on the one or more first function display regions at an angle of 50° to 72°, and the one or more first function display regions have a reflectivity, for the projection light ray for forming the first image, greater than or equal to 4%; and a projection light ray for forming the second image is incident on the one or more second function display regions at an angle of 50° to 72°, and the one or more second function display regions have a reflectivity, for the projection light ray for forming the second image, greater than or equal to 8%.
14. The laminated glass of claim 11, further comprising a dielectric film, wherein the dielectric film

is at least located in the one or more second function display regions.

15. The laminated glass of claim 14, wherein the dielectric film is further located in the one or more first function display regions.

16. The laminated glass of claim 14, wherein the adhesive film is of uniform thickness, a proportion of P-polarized light in the projection light ray for forming the second image ranges from 60% to 100%, the dielectric film is a laminated structure formed by a high refractive index layer and a low refractive index layer stacked with each other or comprises at least one metal layer or laminated PET, and the one or more second function display regions have the reflectivity, for the projection light ray for forming the second image incident at an angle of 50° to 72°, greater than or equal to 10%.

17. The laminated glass of claim 14, wherein the adhesive film is of uniform thickness or wedge-shaped, the dielectric film is disposed on the fourth surface and is an anti-reflective film, and the second function display region is part of the first surface; a proportion of S-polarized light in the projection light ray for forming the second image ranges from 60% to 100%, and the anti-reflective film has a reflectivity for the projection light ray for forming the second image less than or equal to 6%; and the one or more second function display regions have the reflectivity, for the projection light ray for forming the second image incident at an angle of 50° to 72°, greater than or equal to 8%.

18. The laminated glass of claim 14, wherein the adhesive film is wedge-shaped, a proportion of S-polarized light in the projection light ray for forming the second image ranges from 60% to 100%; the dielectric film is a laminated structure disposed on the third surface or the fourth surface and formed by a high refractive index layer and a low refractive index layer stacked with each other; and the one or more second function display regions have the reflectivity, for the projection light ray for forming the second image incident at an angle of 50° to 72°, greater than or equal to 28%.

19. The laminated glass of claim 13, wherein the adhesive film is wedge-shaped, the second function display region is the fourth surface, a proportion of S-polarized light in the projection light ray for forming the second image ranges from 60% to 100%, and the one or more second function display regions have the reflectivity, for the projection light ray for forming the second image incident at the angle of 50° to 72°, greater than or equal to 8%.

20. A head-up display system, comprising a first projection light source and a laminated glass, the laminated glass comprising a first transparent substrate, a second transparent substrate, and an adhesive film, wherein the first transparent substrate has a first surface and a second surface opposite the first surface; the second transparent substrate has a third surface and a fourth surface opposite the third surface, and the third surface is closer to the second surface than the fourth surface; the adhesive film is disposed between the second surface and the third surface and configured to adhere the first transparent substrate and the second transparent substrate; the laminated glass has a light-transmitting region and a light-blocking region surrounding at least part of a periphery of the light-transmitting region; the light-transmitting region has a visible light transmittance greater than or equal to 70%, the light-blocking region has a visible light transmittance less than or equal to 5%, the light-blocking region has a first region located below the light-transmitting region, and the first region has one or more first function display regions for displaying of a first image; and the first projection light source is configured to emit a projection light ray for forming the first image to the first function display region.